CDE2501 SUSTAINABLE SYSTEMS FOR LIVEABLE CITIES

STUDIO GUIDE
AY 2025 – 2026 SEMESTER 2

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OVERVIEW

CDE2501 Sustainable Systems for Liveable Cities introduces the application of fundamental systems thinking concepts through the lens of urban sustainability and liveability in Singapore. Using Singapore Liveability Framework as a guiding principle, the course discusses how historical, socio-economic, environmental factors (among others) shape urban systems (e.g., transportation, housing, economy) and their interactions differently in driving sustainability and liveability trajectories in Singapore. Through case studies, the course examines how approaches to integrated planning, dynamic governance, and a collaborative ecosystem, along with associated decision making and solution development, contribute to sustainability and liveability in Singapore. This examination hinges on interdisciplinary collaboration and trade-offs that involve multiple actors, stakeholders and communities.

By the end of this course, you will be able to:

- Appreciate the systems approach taken by Singapore to achieve and sustain liveability outcomes as described in the Liveability Framework developed by the Centre for Liveable Cities
- Understand the considerations taken (for solving complex challenges faced in the development of a liveable city) to develop initiatives, programmes and interventions needed to address problems and/or make improvements in Singapore.
- Apply systems thinking fundamentals to identify & analyse city problems/challenges, propose, evaluate & recommend solution(s)
- Develop teamwork and communication skills through cross-discipline collaboration

Co-Course Leaders:

1) **Associate Professor Wee Hiang Koon:** Founding and Principal Architect at SKEW Collaborative; Registered Architect
2) **Adjunct Professor Looi Teik Soon:** Advisor and Former Dean, LTA Academy & Singapore Rail Academy; Former Director Policy, LTA
3) **Dr. Rosita Samsudin:** Lecturer, NUS Cities

Course Coordinators (*CC all if you have queries***):**

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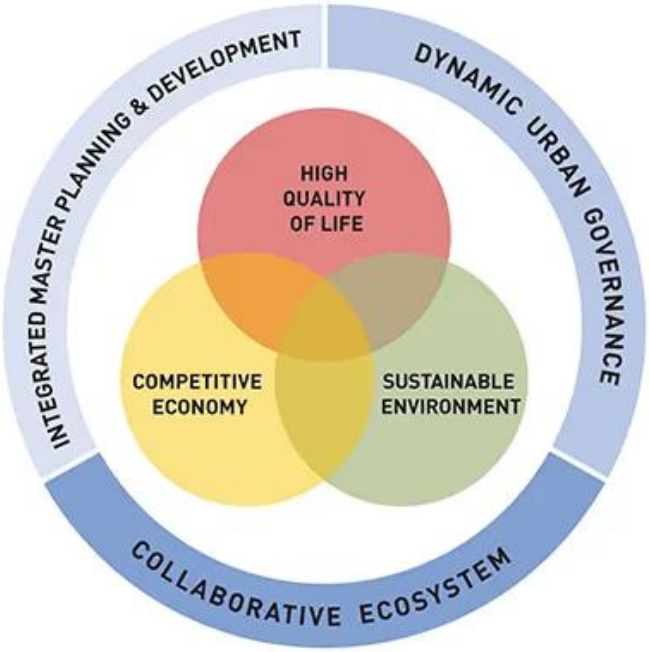
LIVEABILITY FRAMEWORK: CENTRE FOR LIVEABLE CITES (CLC)



Since its independence in 1965, Singapore has developed into a highly liveable city. The Centre for Liveable Cities (CLC) analysed case studies of Singapore's urban development to distil their lessons on what makes a city liveable, and how this liveability can be achieved. The result of these efforts was the Liveability Framework.

The Liveability Framework consists of three key Liveability Outcomes (High Quality of Life, Competitive Economy and Sustainable Environment), which are achieved through three key underlying Systems (Integrated Master Planning and Development, Dynamic Urban Governance, Collaborative Ecosystem).

The Three Liveability Outcomes

High Quality of Life consists of a good mix and balance of:

- Accessible, adequate and affordable public goods and services (housing, transport, water, sanitation, health, etc.)
- Inclusivity, social cohesion, well-being, mental health and sense of belonging (intergenerational matters, heritage, etc.)
- Resilience and digital connectivity

Sustainable Environment consists of a good mix and balance of:

- Clean, good quality environment for immediate and long-term resource needs (clean air; clean and available water; functioning drainage; waste and sewerage systems; control of noise pollution; energy resilience; etc.)
- Holistic green, blue and nature-based eco-systems (habitat, flora and fauna, nature reserves, etc.)
- Resilient to risks (climate change, natural disasters, food security, etc.)

Competitive Economy consists of a good mix and balance of:

- Competitive workforce: jobs, skills and employability
- Attract talents, conducive business environment and inclusive work culture
- Resilient economic growth

The Three Underlying Systems

Integrated Master Planning and Development

This refers to how urban planning and development (to define in a simplified way, the process of developing and designing land use and the built environment) needs to be ***integrated*** because the city is complex (made up of many urban systems e.g. housing, water, nature, arts) and thus many factors need to be considered together when planning and developing the city. This System consists of the following 5 principles:

1) **Think Long Term:** Urban planning and development need to ***integrate*** considering *both* current and future needs of the city and its citizens.
    a. E.g. Singapore has an integrated urban planning system that both plans for the long-term and allows for regular reviews – see below.

Source: Adapted by Looi TS based on URA info, 2025

2) **Build for Resilience:** Planning ahead for the long-term and sticking with these plans is not enough. As the city is complex and lives in a complex world, circumstances and conditions can change, whether gradually or suddenly. Urban planning and development need to be nimble enough to anticipate and respond to these changes, ***integrating*** both planning ahead, and managing and dealing with potential uncertainties.
    a. E.g. Singapore's urban planning system allows for regular reviews of both the Master Plan and Long-Term Plan based on ground realities; government agencies also regularly review their policies – see above.

3) **Seek "Win-Win":** No single government or planning agency has all the knowledge and expertise needed to tackle each urban problem. Urban problems are complex and require the combined expertise of people and institutions with different specialisations. Urban planning and development need to ***integrate*** the knowledge and expertise of different agencies to adequately address urban problems
    a. E.g. Singapore's Green Plan is collectively spearheaded by 5 ministries – the Ministry of Education (MOE), the Ministry of National Development (MND), the Ministry of Sustainability and the Environment (MSE), Ministry of Trade and Industry (MIT), and Ministry of Transport (MOT). This is because the key pillars of the Plan - City in Nature, Sustainable Living, Green Economy, Energy Reset, Resilient Future – have too large of a scope to be addressed by one ministry alone.

4) **Execute Effectively:** Combining the knowledge and expertise of different government and planning agencies to come up with ideas for solutions is not enough. Ideas need be translated into solutions that are achievable in real life. In order to do this, government and planning agencies need to ***integrate*** to coordinate their efforts as they implement different parts of a solution.
    a. E.g. LTA has worked with agencies like the URA, HDB and NParks to align key transport networks, including MRT networks, with different land uses, such as housing and parks. This ensures that transport networks serve the needs of making such land uses accessible to people.

5) **Innovate Systemically:** Agencies cannot be narrow minded. They must innovate and think beyond their usual scope of work to solve urban issues creatively. They must ***integrate*** their area of expertise and familiarity with new ideas.
    a. E.g. PUB had to think beyond the usual approach of using concrete engineering and more extensive drainage systems to manage flooding, enlisting more nature-based solutions to develop Active, Beautiful and Clean (ABC) Waters projects such as Bishan-Ang Mo Kio Park to achieve

better results that also had other pay offs, such as providing new recreational spaces and increasing public engagement with water issues.

Dynamic Urban Governance

Urban governance is defined as the processes through which government is organized and delivered in urban areas, focusing on the relationships between state agencies and civil society, including citizens, communities, and private-sector actors. Urban governance needs to be ***dynamic*** to be adaptable and responsive to challenges that may arise and which may not be expected. This System consists of the following 5 principles:

1) **Lead with Vision and Pragmatism:** Government needs to have a long-term strategic ***vision*** to guide its policies so that they are not short-sighted and reactive to what happens. Government also needs to be ***pragmatic*** – being willing to test and try what works in order to achieve liveability for the people – this means not being held back by traditional ways of doing things if they do not work. This is ***dynamic*** because it ensures that the government is nimble enough to change policies and approaches if the current ones are not working.
    a. E.g. At a time when it was still risky to build a container terminal, Singapore did it in 1969, thinking with a long-term strategic vision, and anticipating that containerisation and demand along the Europe-Far East Route would increase in the future. Singapore was proved right when it overtook Rotterdam in 1982 as the world's busiest port.

2) **Uphold a Culture of Integrity:** Government needs to build trust with the public through being transparent with them about the impacts of its policies. This can be further built through citizen engagement. In addition, it needs safeguards against corruption so that this trust is not broken down.
    a. E.g. the Corrupt Practices Investigation Bureau (CPIB) was established to pursue all instances of corruption, regardless of the sums involved or the prominence of the individuals implicated.

3) **Cultivate Sound Institutions:** Individual decision-makers are prone to biases and personal blind spots. To prevent this, institutions (e.g. government agencies, ministries, cross-agency institutions) should be built that can ensure that policies and decisions are not based only on the perspectives of individuals. Institutions combine the collective wisdom of many people and also have institutional cultures and processes that can serve as checks on the decisions that any one person makes in the organisation. When there is more than one institution, they can serve as checks one another as well. This is ***dynamic*** as this allows decisions

and policies to be changed after rigorous processes of engagement and review by other institutions, and within the institutions themselves.

   a. E.g. The government has ensured that "productive fights" arising from the policy debates result in consensus based on pragmatism such as in the construction of the Bukit Timah Expressway (BKE). Ministry of Sustainability and the Environment (MSE) raised concerns about the BKE's environmental impact. Ministry of National Development (MND) proposed measures to minimise this. This resulted in the creation of Eco-Link@BKE, an ecological corridor allowing animals to move between Bukit Timah and Central Catchment Nature Reserves unimpeded.

4) **Engage and Partner with the Community:** Even the experts in government agencies do not have all the answers. The community experiences realities on the ground directly, and so can possess their own wisdom, knowledge and expertise about these issues. Authorities need to tap onto their knowledge and feedback to better craft policies and interventions that serve the needs and aspirations of the community. This is ***dynamic*** because this allows decisions and policies to be changed after a process of engagement and review with the public to ensure that their needs and aspirations are met.

   a. E.g. To create a shared vision for the Rail Corridor, URA sought ideas from diverse segments of society, including interest groups, nature and heritage enthusiasts, sports associations and event organisers, as well as a wide spectrum of local residents, including the elderly, families and the physical challenges.

5) **Work with Markets:** Government also needs to introduce ***dynamism*** by allowing decisions and policies to be more responsive to productive market forces by engaging the private sector.

   a. E.g. The PUB pioneered the PPP (public-private partnership) model in Singapore, starting with a seawater desalination plant in Tuas in 2005. The PPP approach offers value by ensuring cost-effective operations, promoting technological innovation and contributing to the growth of Singapore's private water industry.

Collaborative Ecosystem

The government cannot make the city liveable on its own. It needs the collective wisdom of partners in the community, the private sector and overseas to achieve this. This System consists of the following 2 principles:

1) **Collective Stewardship as a Foundation for Transformative Collaborations:** Collective stewardship encourages ownership among the community and the private sector for how the city develops. This is encouraged by the government collaborations and engagements with the community and the private sector.
    a. E.g. Community and business groups can partner with URA to set up Business Improvement Districts (BIDs) to enhance visitor experience in their areas. Stakeholders develop detailed business plans to transform their precincts and the surrounding public spaces. Provided they have at least 51% support from stakeholders, the government matches dollar-for-dollar the membership fees collected by each BID up to a certain cap and a certain period.

2) **Forging Strong Partnerships Beyond a City's Boundaries:** there is also a need to collaborate with other countries and regions to draw on each other's strengths
    a. E.g. The Lao PDR-Thailand-Malaysia-Singapore Power Integration Project (LTMS-PIP) is a pathfinder project aimed at enhancing regional power connectivity and energy security in Southeast Asia. It involves the utilisation of existing cross-border transmission infrastructure to facilitate the export of hydropower from Lao PDR to Singapore through Thailand and Malaysia.

Links Between the Liveability Framework & CDE2501

The course is structured around seminars and studios. Seminars contain both lectures and Q&A components that will help you to delve into what the various parts of the Framework mean, and how they have been applicable in the Singapore context.

The studios bring you through a project which will task you with identifying an urban problem and crafting a proposal on how to address that problem.

- The three **Liveability Outcomes (High Quality of Life, Sustainable Environment, Competitive Economy)** become the standards by which you assess whether the site you are investigating has achieved these outcomes. Problems are the gaps between the ideals of the Liveability Outcomes and the current state of the site.

- The three underlying **Systems (Integrated Master Planning & Development, Dynamic Urban Governance, Collaborative Ecosystem)** are not directly related to your project, as your project will deal with the scale of a neighbourhood or district rather than a whole city. But the following ideas behind these Systems will be applicable to your project:

- **Urban problems are complex due to the multiple causes and systems underlying them and this requires an *integrated* approach**
    - In your projects, you will engage with this through the **tools of systems thinking, specifically the Causal Loop Diagram**. Through these tools, you will understand the root causes and underlying systems behind the problems you observe in your site.
- **Any urban issue has multiple stakeholders, whether from government, private sector, or public, and these need to be considered and engaged in the process of developing solutions – this is both a *dynamic* process** (as proposed solutions have to change with the input/consideration of these stakeholders) **and a *collaborative* process** (capitalising on the knowledge that the community and the private sector have about the realities on the ground)
    - In your projects, you will engage with this during the stages of understanding the problem, coming up with solutions, and evaluating your solutions:
        - **Understanding the problem:** Problems are significant because they affect certain stakeholders negatively
        - **Coming up with solutions:** Your solutions are only realisable if you enlist the help of the right stakeholders
        - **Evaluating your solutions:** Solutions will always have both positive and negative impacts on stakeholders, and these vary from stakeholder to stakeholder. Solutions need to be refined and improved so that positive impacts are increased across a wide range of stakeholders, and negative impacts are decreased for the same

For more information about the Liveability Framework, please read: Building Liveable and Sustainable Cities: A Framework for the Future | Centre for Liveable Cities Knowledge Hub

STUDIO PROJECT

The studio project is your main task in this course. The project will give you the chance to identify a real-world urban problem in an allocated site, and craft solutions that can address the problem, giving you a taste of what it is like for planners, policymakers, engineers, architects, designers, community organisers, and more as they intervene in the city to improve its liveability.

Studio Project Process

Step 1: Conduct Site Research and Observation

Research about your site, observe the site, and learn from stakeholders about what are the issues they face. This will enable you to understand the site context and issues in the site.

Step 2: Identify Problem Statement

From your research, observations, interviews and surveys, narrow down to a real-life problem that you want to focus on in your project and attempt to address. You will compare your site's current conditions with the ideals of the Liveability Outcomes (High Quality of Life, Sustainable Environment, Competitive Economy) to identify what are the gaps between the current state and the ideal – these are problems that you can consider. The problem you then select should be expressed in a clear and concise statement that will help to guide the rest of your project.

Step 3: Understand Root Causes and Systems

Once your Problem Statement is identified, you will use systems thinking tools like the Causal Loop Diagram to understand what is causing the problem. These causes can come from a variety of urban systems (e.g. housing, water, health, arts). **This step is important because later on, when you are thinking about how to solve the problem, you need to craft solutions that do not just address the problem at the surface level symptoms, but also at the deeper, underlying causes.**

Step 4: Craft Solution(s)

Craft solution(s) that address the problem at its deeper, underlying causes.

Step 5: Evaluate Solution(s)

Evaluate your solution(s) to see what their positive and negative impacts are, especially for the stakeholders in your site.

Step 6: Refine Solution(s)

Refine your solution(s) to enhance their positive impacts and reduce their negative impacts for a wide range of stakeholders.

Note: these steps are not linear but can be iterative. This means that the findings from a later step can help you to reflect on the earlier steps and change them. E.g. understanding the root causes of the problem you identified may cause you to re-write your Problem Statement to focus on the more fundamental problems rather than their symptoms.

Conduct of Studio Project

Teams

You will be split into teams of 4 to 6 people consisting of students from different majors across the College of Design and Engineering (CDE) and NUS at large. This will give you a taste of what it is like to work in interdisciplinary teams. Just as in the Liveability Framework, where agencies with different expertise need to work together, and work with the community and the private sector to address complex urban problems, so too in your projects you will need the strengths of your teammates to craft good solutions. No one has all the answers.

Sites

The sites will be allocated to you based on a system of balloting. This will be elaborated further when you start your studios. Your team will be allocated one of the following sites:

- Jurong
- Queenstown
- Woodlands
- Tampines
- Chinatown
- Bras Basah-Bugis
- Punggol

Suggested site boundaries are given in the Site Guides which can be found on Canvas. Please note that the site boundaries are only suggested - if there is an area of interest nearby that you would like to tackle, you may do so. However, please consult your Studio Leader and Teaching Assistant on this.

Site Guides also contain prompting questions and starter references (online resources) that can be good starting points for your investigation.

Studio Project Deliverables

The outcome of the studio project will be mainly Presentation 1 (in Studio 4; Week 7) and Final Presentation (in Studio 7; Week 13):

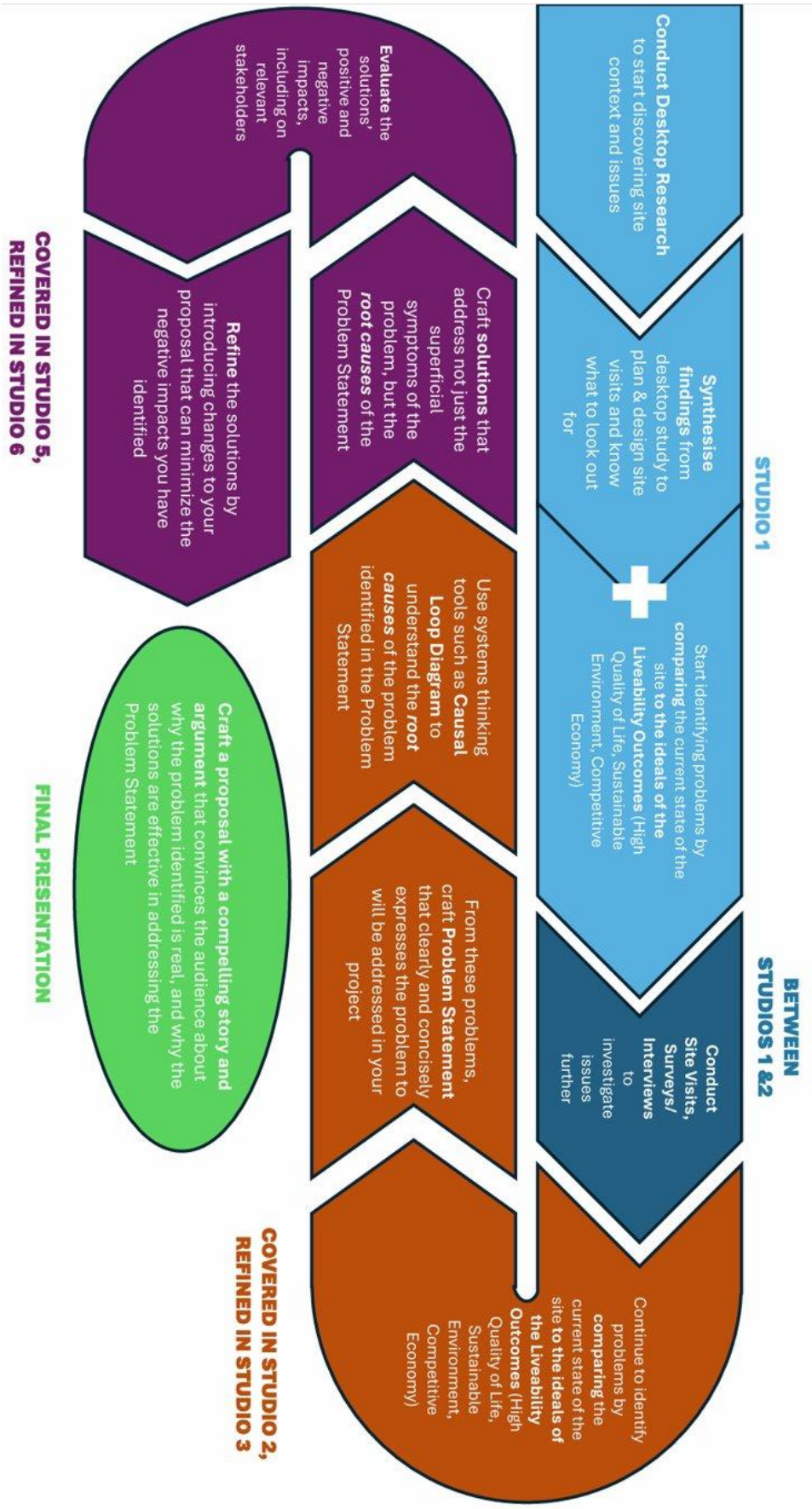
Presentation 1 (Studio 4; Week 7)

- Demonstrate understanding of the **demographics and context of the site**
- **Identify problem(s)** for the given site, to be summarized through a **Problem Statement**
- **Analyse the site and the problem(s) to show root causes and relationships between them**
- **Describe relevant stakeholders and communities** connected to the identified problem
- **Give relevant evidence** to support problem identified

Final Presentation (Studio 7; Week 13)

- **Brief demographics and context of the site** (specific to the problem)
- **Identification of the problem(s)** for the given site, and summarizing it through the **Problem Statement**
- **Analysis of the site and problem(s)** through tools like Causal Loop Diagram
- **Describe the relevant stakeholders/communities** connected to the identified problem.
- **Solution(s)** addressing the root causes of the problem
- **Evaluation of the possible impacts** on the solution(s) on the relevant stakeholder/communities and the possible trade-offs using tools such as S.E.E., BCA, Leopold's Matrix
- **Conclusion**

Studio Process



How The Structure of CDE2501 Supports You in Your Studio Project

Seminars cover the Liveability Framework in more detail and teach skills that will be essential for your project

Studios consist of Consultation Studios and Presentation Studios:

- **Consultation Studios** are where additional content and skills essential for your project will be taught. There will also be time for consultations with your Studio Leader and Teaching Assistant to enable you to refine your project.
- **Presentation Studios** are where you will give your graded presentations and be assessed based on them.

More will be covered on the relationship between seminars, studios, and your project in the next section.

COURSE SCHEDULE AND FLOW

Overview

Week/Dates	Studio/Seminar	Topic
Week 1: 16 Jan 2026 (Fri)	Seminar 1	• Introduction to the Liveability Framework
Week 2: 23 Jan 2026 (Fri)	Seminar 2	• High Quality of Life • Site Context and Observation
Week 3: 27 Jan 2026 (Tue) – 30 Jan (Fri)	Studio 1 (Introduction)	• Site Research and Observation • Surveys and Interviews • Introduction to Problem Statement
Week 4: 6 Feb 2026 (Fri)	Seminar 3	• Sustainable Environment • Systems Thinking and the Causal Loop Diagram
Week 5: 10 Feb 2026 (Tue) – 13 Feb 2026 (Fri)	Studio 2 (Consultation)	• Gathering Findings • Identifying Problem Statement
Week 6: 17 Feb 2026 (Tue) – 20 Feb 2026 (Fri)	Studio 3 (Consultation)	• Refining Causal Loop Diagram • Visualisation Methods • Presentation Tips
Recess Week		
Week 7: 3 Mar 2026 (Tue) – 6 Mar 2026 (Fri)	Studio 4 (Presentation)	See deliverables above
Week 8: 13 Mar 2026 (Fri)	Seminar 4	• Competitive Economy • Crafting Solutions
Week 9: 20 Mar 2026 (Fri)	Seminar 5	• Urban Governance • Evaluating Solutions
Week 10: 24 Mar 2026 (Tue) – 27 Mar 2026 (Fri)	Studio 5 (Consultation)	• Additional Pointers on Crafting Solutions and Evaluating Solutions

Week 11: **Alternative date/time to be confirmed – 3 Apr 2026 (Fri) clashes with Good Friday holiday**	Seminar 6	• The Future of Cities and You
Week 12: 7 Apr 2026 (Tue) – 10 Apr 2026 (Fri)	Studio 6 (Consultation)	• Consultations
Week 13: 14 Apr 2026 (Tue) – 17 Apr 2026 (Fri)	Studio 7 (Presentation)	See deliverables above

Week 1: 16 Jan 2026 (Fri), ***Seminar 1: Introduction to the Liveability Framework***

Lecturer: Adjunct Professor **Looi Teik Soon** (Co-Course Leader, CDE2501 Sustainable Systems for Liveable Cities; Advisor and Former Dean, LTA Academy & Singapore Rail Academy; Former Director (Policy), LTA)

Description: This seminar will introduce you to the **Liveability Framework** and its components (the three Liveability Outcomes – High Quality of Life, Sustainable Environment, Competitive Economy; and the three underlying Systems – Integrated Master Planning & Development, Dynamic Urban Governance, Collaborative Ecosystem). Adjunct Professor Looi will refer to case studies of transport in Singapore to illustrate. The seminar will also introduce you to what you can expect for the rest of the semester in the course's seminars and studios.

Week 2: 23 Jan 2026 (Fri), ***Seminar 2: High Quality of Life + Site Context and Observation***

Lecturers: Practice Professor **Fong Chun Wah** (Deputy Director and Director (Education), NUS Cities; Former Deputy CEO (Building), HDB); Dr. **Rosita Samsudin** (Co-Course Leader, CDE2501 Sustainable Systems for Liveable Cities; Lecturer, NUS Cities; PhD, Architecture, NUS)

Description: This seminar will go into more depth on what a **High Quality of Life** consists of, with reference to how the needs and aspirations of Singapore's citizens have changed over the years, and how Singapore's approach to planning housing and neighbourhoods has evolved with this. This is crucial to understand in your projects as you will be assessing how liveable your site is. Understanding the gap between the ideal of a High

Quality of Life and the current state of your site can help you to start identifying potential issues that you can work on in your project.

This seminar will also touch on what is involved in understanding **Site Context**, and what **Site Observations** you can do to understand your site and its issues better.

Week 3: 27 Jan 2026 (Tue) – 30 Jan (Fri), ***Studio 1 (Introduction)***

Studio Outcomes:

- Recognise how stakeholders, demographics and spatial contexts shape your allocated site.
- Identify and apply appropriate fieldwork strategies (e.g., conducting site visits, surveys and interviews) to investigate your site.

Week 4: 6 Feb 2026 (Fri), ***Seminar 3: Sustainable Environment + Systems Thinking and the Causal Loop Diagram***

Lecturers: Dr. **Rosita Samsudin** (Co-Course Leader, CDE2501 Sustainable Systems for Liveable Cities; Lecturer, NUS Cities; PhD, Architecture, NUS); Dr. **Corinne Ong** (Senior Lecturer, Residential College 4 (RC4))

Description: This seminar will go into more depth on what a **Sustainable Environment** consists of. This is crucial to understand in your projects as you will be assessing how liveable your site is. Understanding the gap between the ideal of a Sustainable Environment and the current state of your site can help you to start identifying potential issues that you can work on in your project.

This seminar will also cover what **systems thinking** is and introduce **the Causal Loop Diagram (CLD)** as a tool to uncover the underlying systems and causes behind an urban problem so that you can apply it to the Problem Statement you will identify.

Week 5: 10 Feb 2026 (Tue) – 13 Feb 2026 (Fri), ***Studio 2 (Consultation)***

Deliverables to Prepare and Bring to Studio:

- Research findings, observations (e.g. in photos, words), results of interviews and surveys. These will be the raw materials from which you will begin to identify issues in your site that may contribute to a Problem Statement
    - You should have done your first site visit before Studio 2!

Studio Outcomes:

- Understand the needs, behaviours, goals and challenges of key stakeholders within your allocated site
- Compare the current state of the site (as observed and researched) to the ideals of the Liveability Outcomes of the Liveability Framework (High Quality of Life, Sustainable Environment, Competitive Economy)
- Apply causal loop diagrams to map and interpret the interconnections, feedback mechanisms and dynamic relationships among urban systems
- Craft a problem statement that reflects (an) issue(s) in your site that your team wishes to tackle for the rest of the project.

Week 6: 17 Feb 2026 (Tue) – 20 Feb 2026 (Fri), ***Studio 3 (Consultation)***

Deliverables to Prepare and Bring to Studio:

- Prepare draft Problem Statement and Causal Loop Diagram for consultation with Studio Leader and Teaching Assistant

Studio Outcomes:

- Refine Problem Statement, Stakeholder Analysis and Causal Loop Diagram based on evidence gathered
- Develop effective presentation techniques to deliver a compelling pitch
- Identify suitable methods and strategies for creating clear and insightful visualization tools

Note: For those whose studios clash with the Chinese New Year Public Holiday (on Tue 17 Feb – Wed 18 Feb), your Studio Leader and Teaching Assistant will liaise with you on alternative timings to hold the studio.

Week 7: 3 Mar 2026 (Tue) – 6 Mar 2026 (Fri) [after Recess Week], ***Studio 4 (Presentation)***

Deliverables for each team's presentation:

- Demonstrate understanding of the **demographics and context of the site**
- **Identify problem(s)** for the given site, to be summarized through a **Problem Statement**
- **Analyze the site and the problem(s) to show root causes and relationships between them**
- **Describe relevant stakeholders and communities** connected to the identified problem
- **Give relevant evidence** to support problem identified

Each team will have to cover these in a **maximum of 12 minutes.**

Week 8: 13 Mar 2026 (Fri), ***Seminar 4: Competitive Economy + Crafting Solutions***

Lecturers: Associate Professor **Wee Hiang Koon** (Co-Course Leader, CDE2501 Sustainable Systems for Liveable Cities; Founding and Principal Architect at SKEW Collaborative; Registered Architect); Mr. **Soh Leng Wan** (Assistant Chief Executive Officer (Manufacturing & Engineering), Enterprise Singapore)

Description: This seminar will go into more depth on what a **Competitive Economy** consists of. In particular, Mr. Leng Wan (Enterprise SG) will be sharing in the seminar about how Singapore's urban economy has evolved to respond to changing conditions in the global economy and in the city. This will be a useful case study to understand what are the considerations that agencies such as Enterprise SG, EDB and URA have when coming up with solutions and interventions in Singapore to ensure it remains economically competitive.

This links to the other topic of this seminar: **Crafting Solutions**. Professor Koon will go through the main considerations you will need to take note of as you craft your solutions to your Problem Statement in your project, so that they address the underlying causes and systems behind the problem, instead of just superficial proposals.

Week 9: 20 Mar 2026 (Fri), ***Seminar 5: Urban Governance & Collaborations + Evaluating Solutions***

Lecturer: Adjunct Professor **Looi Teik Soon** (Co-Course Leader, CDE2501 Sustainable Systems for Liveable Cities; Advisor and Former Dean, LTA Academy & Singapore Rail Academy; Former Director (Policy), LTA); **Guest Speaker** (to be confirmed)

Description:

Case studies of **urban governance and collaborative eco-system**, where government agencies do not make policies and decisions by themselves only, but engage the public, NGOs and private sector to craft solutions that are sensitive to the needs of the various stakeholders. Decisions cannot be made unilaterally, especially with the complexity of the issues being addressed.

This covers **Evaluating Solutions,** using the environmental and economic (SEE) method to assess the impacts on different stakeholder groups. This can be applied to students' proposals for solutions in their studio project to understand the impacts of their proposed solutions.

Week 10: 24 Mar 2026 (Tue) – 27 Mar 2026 (Fri), ***Studio 5 (Consultation)***

Deliverables to Prepare and Bring to Studio:

- First-cut solutions for your Problem Statement
- Preliminary evaluations of your draft solutions

Studio Outcomes:

- Understand that systems solutions are inherently multi-pronged, engage multiple urban systems and address interconnected causes rather than isolated symptoms
- Develop a systems solution that addresses the problem statement and its underlying root causes using insights gathered from site analysis and other research
- Evaluate how the proposed solution addresses the problem statement, its root causes and the interrelationships between urban systems

Week 11: **Alternative date/time to be confirmed – 3 Apr 2026 (Fri) clashes with Good Friday holiday,** **Seminar *6: The Future of Cities and You***

Guest Speakers: To be confirmed

Description: In this seminar, guest speakers from various urban fields, who have come from design and engineering backgrounds, will share more about their career journeys, and their experiences of tackling real-world urban issues. You may come to see how the things you are learning in this course can be relevant to your future career, or as informed citizens.

Week 12: 7 Apr 2026 (Tue) – 10 Apr 2026 (Fri), ***Studio 6 (Consultation)***

Deliverables to Prepare and Bring to Studio:

- Draft of Problem Statement, problem analysis, proposed solutions, evaluation, and how you would structure them in the Final Presentation

Studio Outcomes:

- Make final refinements to your Problem Statement, problem analysis, proposed solutions, evaluation, and presentation flow and structure via consultation with your Studio Leader and Teaching Assistant

Week 13: 14 Apr 2026 (Tue) – 17 Apr 2026 (Fri), ***Studio 7 (Presentation)***

Deliverables for each team's presentation:

- **Brief demographics and context of the site** (specific to the problem)
- **Identification of the problem(s)** for the given site, and summarising it through the **Problem Statement**
- **Analysis of the site and problem(s)** through tools like Causal Loop Diagram
- **Describe the relevant stakeholders/communities** connected to the identified problem.
- **Solution(s)** addressing the root causes of the problem
- **Evaluation of the possible impacts and trade-offs** of the solution(s):
    - **on the relevant stakeholders/communities** (Stakeholder Impact Evaluation) and
    - **on the social, environmental and economic impacts of the solution(s)** [e.g. through S.E.E. Evaluation, Leopold Matrix, Cost-Benefit Analysis or any other evaluation method that shows the social, environmental and economic impacts of the solution(s)]
- **Conclusion**

Each team will have to cover these in a **maximum of 12 minutes.**

COURSE ASSESSMENT

	Grading Component	Weightage (%)
Group	Presentation 1	15
	Final Presentation	25
Individual	Individual Presentation	10
	Individual Contribution & Engagement	15
	Peer Review	10
	Seminar-based Quizzes & Assignments	25

Group Grade (40% Total)

As mentioned earlier in this guide, Teams will do two presentations during the semester. One in Week 7 and one in Week 13.

- **Presentation 1 (15%):** The first presentation will be a maximum of 12 minutes and scored based on a rubric. This rubric can be found on Canvas, titled '**Presentation Rubrics**'.
- **Final Presentation 2 (25%):** The second and final presentation will be a maximum of 12 minutes and scored based on a rubric. This rubric can be found on Canvas, titled '**Presentation Rubrics**'.

One member of the team must submit the presentation slides at **5:00 pm the day before the presentation studio.** (e.g. If your presentation studio is on Tuesday, you must submit the slides on Monday at 5:00 pm).

Please note that the **Studio Leaders and Teaching Assistants will moderate your group grade if more than half the group complains** about a teammate not participating enough in group work, site visits, content creation, ideation, and discussion. The teaching team will consider Peer Reviews, discussions and consultations with students when moderating

Individual Performance (60% Total)

Individual Presentation (10%, 5% For Each Presentation)

You are required to present for both presentations unless you have a valid reason (MC, LOA) for missing presentation studio. You will be graded based on:

- Your ability to speak in a confident, clear, and understandable manner
- Eye contact with the audience instead of with the screen, computer, phone, or script

Individual Contribution & Engagement (15%)

This considers your quality of:

- Contribution to and engagement in group work
- Engagement during group consultations and discussions
- Participation in Question & Answer segments after presentation
- Participation in class discussions

Peer Review Score (10%)

There will be two **COMPULSORY** peer reviews, a mid-semester one by the end of Week 7 and a final one by the end of Week 13. Students will grade their peers based on the rubric which can be found on Canvas, titled '**Peer Review Rubric**'.

The mid-semester peer review must be submitted after the first presentation by Week **7, Friday, 6 March 2026, 11:59pm.** The score will be based on the average that every team member gives their peer. **The teaching team will send a link to the peer review at the end of Recess Week.**

The final peer review must be submitted after the final presentation by **Week 13, Friday, 17 April 2026, 11:59pm.** The score will be based on the average that every team ember gives their peer. **The teaching team will send a link to the peer review at the end of Week 12.**

***NOTE:** These surveys will be strictly confidential, with results only made known to the students' respective teaching team (SL & TA).*

Seminar-based Assignments & Quizzes (25% Total, 4 Online Assignments Worth 6.25% Each)

Seminar-based assignments will assess students' comprehension and application of seminars. They will come in the form of either MCQ quizzes or short reflection exercises.

Two types of seminar-based graded assessment:

- **Quiz**: For each quiz, you will have 8 minutes to complete 5 MCQ questions. Each quiz is worth 5 marks each.
- **Individual assignments**: For each assignment, you will need to complete 1 short answer question. Each assignment is worth 5 marks each.

Each quiz/individual assignment forms 6.25% of your grade. In this course, students will have to complete a total of 2 quizzes and 2 individual assignments.

Each quiz will be held during a particular seminar, and these will test students on the content of the seminar. Individual Assignments will come after Seminar 3 and Seminar 5 respectively and will ask you to reflect on your learnings from the seminars.

Quiz/Assignment	**Date that Quiz/Assignment will be open on Canvas**	**Topic of Quiz/Assignment**
Quiz 1*	Fri 23 Jan (During Seminar 2)	Seminar 2
Quiz 2*	Fri 6 Feb (During Seminar 3)	Seminar 3
Quiz 3*	Fri 13 Mar (During Seminar 4)	Seminar 4
Quiz 4*	Fri 20 Mar (During Seminar 5)	Seminar 5
Individual Assignment 1	Fri 6 Feb – Fri 13 Feb (due 11:59pm)	Reflection on Seminars 1-3
Individual Assignment 2	Fri 20 Mar – Fri 27 Mar (due 11:59pm)	Reflection on Seminars 4-5

***IMPORTANT - Note on Quizzes:**

- You will be allocated to attend **2 seminars in person** (among Seminars 2 to 5)
- **For the seminars you are allocated**, you **have to** do the quizzes that will be conducted during the seminars
- **But for the seminars you are not allocated,** you **do not** have to do the quizzes that will be conducted during the seminars

- Quiz password will be only made visible to those in the lecture theatre, so **only those attending in person can do the quiz**
- By the end of the course, **you should have completed 2 out of 4 of the quizzes**

LATE SUBMISSIONS

Studio Project

Students should submit presentation materials to Canvas by **5:00 pm the day before the presentation studio** (i.e., submit by 5:00 pm on Monday if your presentation studio falls on Tuesday).

Only one student per group will need to submit their group presentation materials. If students submit late, points from the overall team score from that studio will be deducted. Submissions received within the 10-minute buffer time will not be marked as late, but students will be given a warning. Submissions received after the buffer time will be demoted by one bound in the grading sheet (e.g., if the team would have received a "A-" for the team score, they would be demoted to a "B+").

Individual Assignments and Peer Reviews

Deadline extensions will only be granted to students with genuine issues (e.g. technical disruptions, personal emergencies, family emergencies). Students who forget to submit by the deadline due to personal carelessness would not be considered as having a valid cause for extension. They will get a zero-mark for that particular assignment or peer review.

ACADEMIC MISCONDUCT

Academic honesty and integrity are essential conditions for the pursuit and acquisition of knowledge, and the University expects each student to maintain and uphold the highest standards of integrity and academic honesty at all times. The University takes a strict view of cheating in any form, deceptive fabrication, plagiarism and violation of intellectual property and copyright laws.

Any student who is found to have engaged in such misconduct will be subject to disciplinary action by the University. Any student found to have committed or aided and abetted the offence of plagiarism may be subject to disciplinary action. In addition, the student may receive no grade for the relevant academic assignment, project, or thesis; and he/she may fail or be denied a grade for the relevant subject or module. (Excerpt from Code of Student Conduct).

SUMMARISED COURSE SCHEDULE

Legend:

	Seminar
	Individual Submission Deadline
	Consultation Studio
	Presentation Studio

Week	Date/Time	Studio/Seminar	Topic/Remarks
Week 1:	16 Jan 2026 (Fri) 10:00am – 12:00pm	Seminar 1	• Introduction to the Liveability Framework
Week 2:	23 Jan 2026 (Fri) 10:00am – 12:00pm	Seminar 2	• High Quality of Life • Site Context and Observation
		Quiz 1*	Based on Seminar 2; held during seminar
Week 3:	27 Jan 2026 (Tue) – 30 Jan (Fri) 9:00am – 12:00pm OR 2:00pm – 5:00pm **(Refer to studio allocation for the date/time of your studio, which will be released by the Monday of Week 3 – 26 Jan 2026)**	Studio 1 (Introduction)	• Site Research and Observation • Surveys and Interviews • Introduction to Problem Statement
Week 4:	6 Feb 2026 (Fri) 10:00am to 12:00pm	Seminar 3	• Sustainable Environment • Systems Thinking and the Causal Loop Diagram
		Quiz 2*	Based on Seminar 3; held during seminar
	6 Feb 2026 (Fri) 12:00pm	Opening of Individual Assignment 1	On Canvas

Week 5:	10 Feb 2026 (Tue) – 13 Feb 2026 (Fri) 9:00am – 12:00pm OR 2:00pm – 5:00pm	Studio 2 (Consultation)	• Gathering Findings • Identifying Problem Statement
	By 13 Feb 2026 (Fri) 11:59pm	Individual Assignment 1 Due	On Canvas
Week 6:	17 Feb 2026 (Tue) – 20 Feb 2026 (Fri) 9:00am – 12:00pm OR 2:00pm – 5:00pm	Studio 3 (Consultation)	• Refining Causal Loop Diagram • Visualisation Methods • Presentation Tips
	20 Feb 2026 (Fri) 8:00am	Opening of Mid-Semester Peer Review	TEAMMATES link will be sent to your email
Recess Week			
Week 7:	3 Mar 2026 (Tue) – 6 Mar 2026 (Fri) 9:00am – 12:00pm OR 2:00pm – 5:00pm	Studio 4 (Presentation)	See deliverables above
	By 6 Mar 2026 (Fri) By 11:59pm	Mid-Semester Peer Review Due	On TEAMMATES
Week 8:	13 Mar 2026 (Fri) 10:00am – 12:00pm	Seminar 4	• Competitive Economy • Crafting Solutions
		Quiz 3*	Based on Seminar 4; held during seminar
Week 9:	20 Mar 2026 (Fri) 10:00am – 12:00pm	Seminar 5	• Urban Governance • Evaluating Solutions
		Quiz 4*	Based on Seminar 4; held during seminar
	12:00pm onwards	Individual Assignment 2 Opens	On Canvas
Week 10:	24 Mar 2026 (Tue) – 27 Mar 2026 (Fri) 9:00am – 12:00pm	Studio 5 (Consultation)	• Additional Pointers on Crafting Solutions and

	OR 2:00pm – 5:00pm		Evaluating Solutions
	By 27 Mar 2026 (Fri) 11:59pm	Individual Assignment 2 Due	On Canvas
Week 11:	**Alternative date/time to be confirmed – 3 Apr 2026 (Fri) clashes with Good Friday holiday**	Seminar 6	• The Future of Cities and You
	2 Apr 2026 (Thu) 8:00am	Opening of End-Semester Peer Review	TEAMMATES link will be sent to your email
Week 12:	7 Apr 2026 (Tue) – 10 Apr 2026 (Fri) 9:00am – 12:00pm OR 2:00pm – 5:00pm	Studio 6 (Consultation)	• Consultations
Week 13:	14 Apr 2026 (Tue) – 17 Apr 2026 (Fri) 9:00am – 12:00pm OR 2:00pm – 5:00pm	Studio 7 (Presentation)	See deliverables above
	By 17 Apr 2026 (Fri) By 11:59pm	End-Semester Peer Review Due	On TEAMMATES

***IMPORTANT - Note on Quizzes:**

- You will be allocated to attend **2 seminars in person** (among Seminars 2 to 5)
- **For the seminars you are allocated**, you **have to** do the quizzes that will be conducted during the seminars
- **But for the seminars you are not allocated,** you **do not** have to do the quizzes that will be conducted during the seminars
- Quiz password will be only made visible to those in the lecture theatre, so **only those attending in person can do the quiz**
- By the end of the course, **you should have completed 2 out of 4 of the quizzes**